

SE 4458 Software Architecture & Design of Modern Large Scale Systems - Final

Group 1: Create a Hotel Booking System

Create a hotel booking system like Hotels.com. Implement Below are use cases for functional and non-functional requirements.

HOTEL ADMIN SERVICE



The hand-drawn UI for the Hotel Swiss Admin Service includes the following elements:

- Header:** "Hotel Swiss"
- Baslangic (Start):** A date input field with slashes (//) and a calendar icon.
- Bitis (End):** A date input field with slashes (//) and a calendar icon.
- Oda Tipi (Room Type):** A dropdown menu with "Standard, Aile" and a downward arrow.
- Oda Adedi (Room Count):** A text input field.
- Radio Buttons:** Two options, "Dolu" (Occupied) and "Bos" (Empty), with the "Bos" option selected.
- Action Button:** A button labeled "DUZELT" (Correct).

- Admins for hotels can add/update rooms for availability between start and end dates
- Image uploading is not necessary but nice-to-have
- THIS WILL BE AN AUTHENTICATED SERVICE

HOTEL SEARCH SERVICE



The screenshot shows the Hotels.com search interface with the following details:

- Header:** "Hotels.com" logo, "Seyahat Ara" (Travel Search) button, and "Baris" text.
- Section Header:** "Nereye?" (Where?)
- Search Fields:**
 - Varis noktası (Destination):** "Bodrum, Muğla, Türkiye"
 - Tarihler (Dates):** "12 May - 16 May"
 - Misafir sayısı (Number of guests):** "2 misafir, 1 oda" (2 guests, 1 room)
- Action Button:** A blue button labeled "Ara" (Search).

Haritada göster

300 veya daha fazla konaklama yeri

Konaklama yeri sıralamamızın işleyiş şekli

Sıralama ölçütü:

Önerilen

Giriş yaptığınızda binlerce otelde ortalama %15 oranında kazanç sağlarsınız

Giriş yap

Hyde Bodrum - Yetişkin Oteli

Torba

Her şey dâhil Havuz

43.791 TL

4 gece için

Gecelik 10.948 TL

vergiler ve ücretler dâhildir

Puan toplayın

10 Olağanüstü 3 yorum

Ekstra kazanç için hesabınıza giriş yapın

MGallery The Bodrum Hotel Yalıkavak

Yalıkavak

Havuz

37.830 TL

4 gece için

Gecelik 9.458 TL

vergiler ve ücretler dâhildir

Puan toplayın

9,6 Olağanüstü 30 yorum

Staff Agent

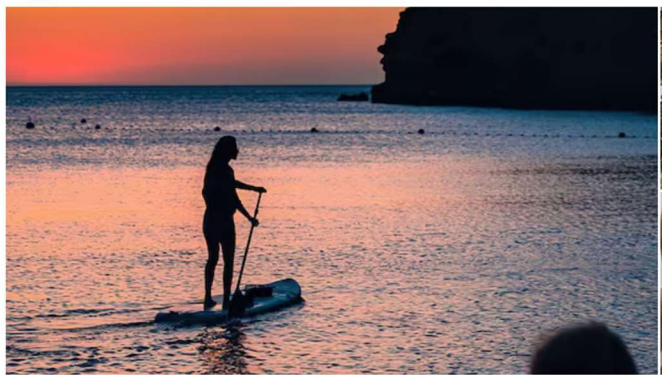
You are chatting with an AI Agent.

Write a message...

Send

- User can search hotels by destination point, dates and number of people. Only rooms that admins specified as vacant will appear in results for given dates
- Client who login to application will see 15% discounted prices.
- 'Haritada goster' is required to show hotels that have been searched

BOOK HOTEL SERVICE



Genel Bakış Otel Özellikleri Odalar Engellilere Yönelik Özellik Politika

Hyde Bodrum - Yetişkin Oteli

8,8 Mükemmel

32 yorumun tümünü göster

Üye Fiyatı: %10 indirim

42.225 TL

~~46.917 TL~~

4 gece için

Gecelik 10.556 TL

vergiler ve ücretler dâhildir

Fiyat bilgileri

Rezervasyon yap

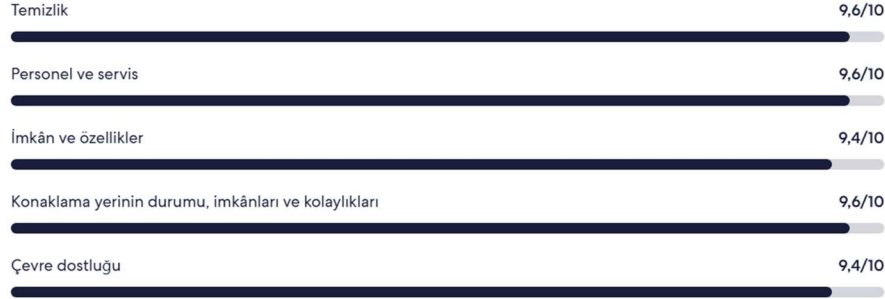
- On hotel detail page, user can book the hotel. Capacity of the hotel needs to be decreased for specified dates
- NO transaction needed for payment

HOTEL COMMENTS SERVICE

- On clicking of comments, a graphs that shows distribution of comments per service will be shown along with comments and ratings

9,6/10 Olağanüstü

163 doğrulanmış yorum ⓘ



8/10 Çok İyi

16 Haziran 2025



Simge, 4 gecelik seyahat

16 Haziran 2025 tarihinde Enver tarafından gönderilen yanıt

Geri bildiriminiz için teşekkür ederiz.

NOTIFICATION SERVICE

- write a nightly scheduled task
 - o to go over all hotel capacities and notify hotel administrators when it is below 20% for the next month.
 - o to pull new hotel reservations from the queue and send them a message about reservation details

AI AGENT SERVICE

Implement AI agent chat window in main application screen that will use APIs you created to perform search and booking use cases

You: I'd like to book a hotel in Rome from July 15 to July 18 for two adults.

AI: Great! Any preferences for hotel rating, price range, or amenities?

You: 4+ stars, breakfast, and a pool would be ideal.

AI: Got it! Here are a few options you might like 🙌

Hotel Roma Plaza

- 📍 City Centre
- ⭐ 4.5 | 💰 \$210/night
- ✅ Free Wi-Fi, Breakfast, Pool

- [Reserve Room]

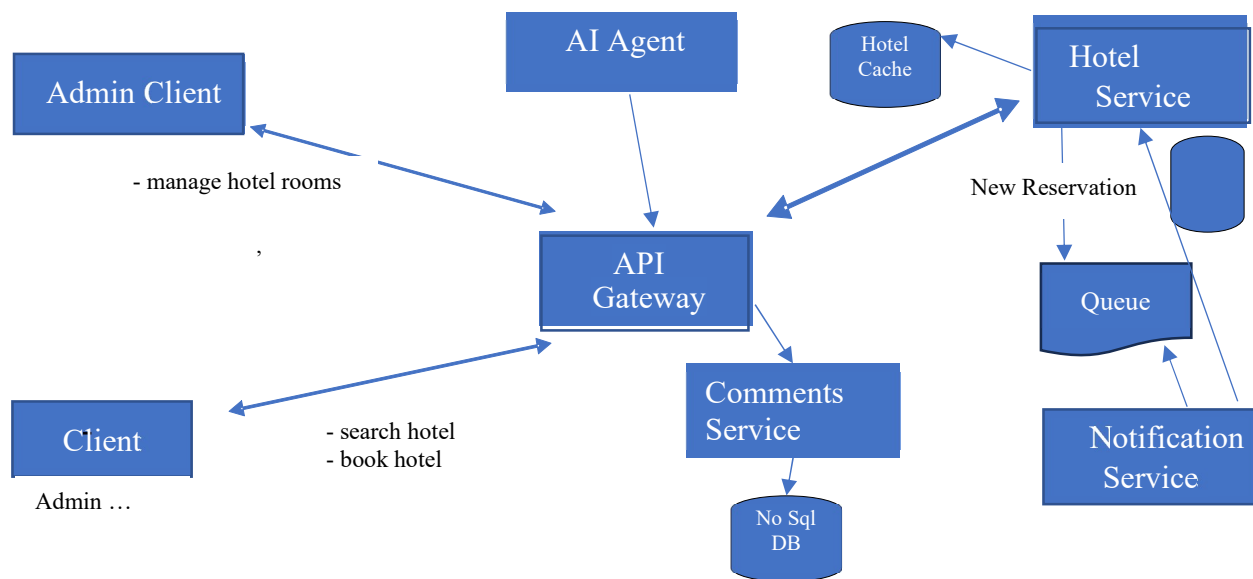
Grand Hotel

- Monti
- ★ 4.3 | 💰 \$250/night
- ✓ Free Wi-Fi, Breakfast, Pool
- [Reserve Room]

AI: Would you like to confirm your reservation at Hotel Roma Plaza from July 15–18 for 2 adults?

○ Yes, book it

NON-FUNCTIONAL REQUIREMENTS



- Comments will be stored in a separate No SQL DB
- Hotel details will be stored in a separate distributed cache like Redis

Students

GİZEM TANIŞ ISIL GULMUS BARTU TASKIN GULCE DOGRUOZ YIGIT ALP YUKSEL ANIL KORKMAZ EDA TERZIOGLU DOGA SACIKARA ÖZGECAN AKTAŞ IŞIL ÜLGER	EFE FENER BİLGEHAN BAYRAMOĞLU KAAAN MERT KOZALI SERKAN ACAR REMZI YAMAN ESEN BERKANT GÜRCAN IŞIL KAHRAMAN BEDİRHAN KIRTAN
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